

Math 350 Section 1
Fall 2024
Exam III
November 19, 2024
Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.
- **Default metric:** unless otherwise stated, the set $\mathbb{R}$ or $\mathbb{R}^n$ will be considered a metric space with the Euclidean metric

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write TRUE if the statement is true. Otherwise, write FALSE. Your response should be in ALL CAPS. No justification is required.

(a) there is a continuous, real-valued function on $[0, 1]$ whose range is $[0, \infty)$

(b) a sequence $\{x_n\}$ of numbers in $\mathbb{R}$ which is monotone increasing and bounded above converges

(c) if $f : (a, b) \rightarrow \mathbb{R}$ is differentiable at a point $c \in (a, b)$, then it is also continuous at c

(d) a Cauchy sequence in $\mathbb{R}$, but with the discrete metric, will converge

(e) the equation $x^{25} + 2x + 2 = 0$ has a solution in the interval $[-1, 0]$

2. (10 points)

(a) Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$. Write down what it means for a function $f : A \rightarrow T$ to be continuous at a point $a \in A$.

(b) If $f : A \rightarrow T$ is a function, write down the definition of the preimage $f^{-1}(V)$ of a subset $V \subseteq T$.

(c) Prove that if $f : A \rightarrow T$ is continuous at every point of A , then for any open subsets $V \subseteq T$, the preimage $f^{-1}(V)$ is open.

3. (10 points)

(a) Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$. If $a \in S$ is accumulation point of A and $f : A \rightarrow T$ is a function, write the definition of $\lim_{x \rightarrow a} f(x) = L$.

(b) Prove that if $\lim_{x \rightarrow a} f(x) = L$, then for any sequence $\{x_n\}$ of elements in A converging to a we have $\lim_{n \rightarrow \infty} f(x_n) = L$

(c) Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

Does the limit $\lim_{x \rightarrow 0} f(x)$ exist? Explain.

4. (10 points)

(a) Write down the definition of a metric space (M, d) being disconnected, as well as the definition of a subset $A \subseteq M$ being disconnected.

(b) Prove that if $A, B \subseteq M$ are connected and overlap, then $A \cup B$ is also connected.

(c) Give an example showing that if $A, B \subseteq M$ are connected and overlap, then $A \cap B$ may not be connected.

5. (10 points)

(a) Let f be a real-valued function on (a, b) . Write down what it means for f to be differentiable at $c \in (a, b)$.

(b) Carefully state the Helper Theorem we used in class.

(c) Use the Helper Theorem to prove the Product Rule for derivatives.

6. (10 points) Let f and α be a real-valued function on an interval $[a, b]$.

(a) Given a partition P of $[a, b]$ and sample points $\{t_k\}$, define $S(P, f, \alpha, \{t_k\})$

(b) Write the definition of the Riemann integral of f on $[a, b]$ with respect to α .

(c) Let $\alpha(x) = x$ and f be the function

$$f(x) = \begin{cases} 0, & x < 1 \\ 1, & 1 \leq x < 2 \\ 2, & 2 \leq x \end{cases}$$

Use the definition to show that f is Riemann integrable with respect to α on $[0, 3]$ and calculate $\int_0^3 f d\alpha$

7. (10 points)

- (a) Use the definition of a derivative to prove that $f(x) = \sqrt{x}$ is differentiable and satisfies $f'(a) = \frac{1}{2\sqrt{a}}$ for all $a > 0$

- (b) Prove that for any $a > 0$

$$\frac{1}{2\sqrt{a+1}} < \sqrt{a+1} - \sqrt{a} < \frac{1}{2\sqrt{a}}$$

Hint: think about the Mean Value Theorem for $f(x) = \sqrt{x}$